

Practice Questions for 2024-11-14-NetAppls

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1 Questions

1. In the transportation problem with inequality constraints (Slides 1-2), if the total supply $\sum_i s_i$ exceeds the total demand $\sum_j d_j$, what modification is necessary to transform the problem into a standard transportation problem solvable with equality constraints, and why is this modification needed?
 - a. Introduce artificial variables to the supply constraints.
 - b. Introduce a dummy destination node with a demand equal to the difference between total supply and total demand.
 - c. Introduce a dummy source node with a supply equal to the difference between total demand and total supply.
 - d. Relax the equality constraints to inequalities.
 - e. Adjust the supply and demand values proportionally to achieve balance.
2. Slides 3 and 4 introduce a dummy node to handle inequality constraints in the transportation problem. Explain the role of the additional constraint derived in Slide 4, $\sum_i s_i - \sum_i x_{id} = \sum_j d_j + \sum_j x_{dj}$, in ensuring the feasibility and solvability of the modified problem.
 - a. It ensures that the total flow into the dummy node equals the total flow out of the dummy node.
 - b. It guarantees that the total supply equals the total demand, including the flow to and from the dummy node.
 - c. It enforces non-negativity of the flows to and from the dummy node.
 - d. It ensures that the objective function remains linear after the introduction of the dummy node.
 - e. It is not necessary for feasibility or solvability; it is simply a consequence of the other constraints.
3. In Slide 5, a transportation problem with unbalanced supply and demand is transformed by introducing a dummy node. Explain the economic interpretation of the zero-cost arcs connecting the dummy node to the sources and sinks.
 - a. They represent the cost of storing excess supply or procuring additional supply to meet demand.
 - b. They represent penalties for unmet supply or demand.
 - c. They represent the cost of transporting goods between sources and sinks.
 - d. They represent the opportunity cost of unused resources.
 - e. They represent a way to mathematically balance supply and demand without incurring any additional cost.

4. Slides 7-8 present a more general approach to handling inequality constraints in network flow problems using a dummy node. Explain how this approach differs from the specific method used for the transportation problem in Slides 3-5, and what advantages or disadvantages this more general approach might offer.
 - a. The general approach uses artificial variables, while the transportation problem approach uses slack variables.
 - b. The general approach is limited to problems with only inequality constraints, while the transportation problem approach can handle both equality and inequality constraints.
 - c. The general approach uses a single dummy node to handle both excess supply and unmet demand, while the transportation problem approach may use separate dummy nodes for each.
 - d. The general approach is more flexible and can be applied to a wider range of network flow problems with various types of inequality constraints, while the transportation problem approach is specific to transportation problems.
 - e. The general approach is computationally less efficient than the transportation problem approach.
5. In the transportation problem with inequality constraints (Slides 1-2), if the total supply $\sum_i s_i$ is less than the total demand $\sum_j d_j$, how can the problem be modified to ensure feasibility and solvability using standard transportation problem algorithms?
 - a. Introduce a dummy source node with a supply equal to $\sum_j d_j - \sum_i s_i$ and zero-cost arcs connecting it to all demand nodes.
 - b. Increase the supply of each source node proportionally to match the total demand.
 - c. Decrease the demand of each demand node proportionally to match the total supply.
 - d. Ignore the inequality constraints and solve the problem as if it were a standard transportation problem with equality constraints.
 - e. Introduce a dummy demand node with a demand equal to $\sum_i s_i - \sum_j d_j$ and zero-cost arcs connecting it to all supply nodes.
6. In the transportation problem with inequality constraints (Slide 1), if the total supply $\sum_i s_i$ is less than the total demand $\sum_j d_j$, how can the problem be modified to ensure feasibility and solvability using standard transportation problem algorithms?
 - a. Introduce a dummy source with a supply equal to $\sum_j d_j - \sum_i s_i$ and zero-cost arcs to all destinations.
 - b. Reduce the demands d_j proportionally until $\sum_j d_j = \sum_i s_i$.
 - c. Ignore the excess demand and solve the problem with the available supply.
 - d. Introduce a dummy destination with a demand equal to $\sum_i s_i - \sum_j d_j$ and zero-cost arcs from all sources.
 - e. Increase the supplies s_i proportionally until $\sum_i s_i = \sum_j d_j$.
7. Slides 3 and 4 introduce a dummy node to handle inequality constraints in the transportation problem. Explain the role of the additional constraint derived in Slide 4, $\sum_i s_i - \sum_i x_{id} = \sum_j d_j + \sum_j x_{dj}$, in ensuring the feasibility and solvability of the modified problem.
 - a. The constraint ensures that the total cost of transportation remains minimized.

- b. The constraint ensures that the total supply equals the total demand in the modified problem.
 - c. The constraint ensures that the flow on each arc is non-negative.
 - d. The constraint ensures that the supply at each source node is not exceeded.
 - e. The constraint ensures that the demand at each destination node is met exactly.
8. In Slide 5, a transportation problem with unbalanced supply and demand is transformed by introducing a dummy node. Explain the economic interpretation of the zero-cost arcs connecting the dummy node to the sources and sinks.
- a. These arcs represent the cost of storing excess supply or procuring additional supply to meet demand.
 - b. These arcs represent penalties for unmet supply or demand.
 - c. These arcs represent the cost of transporting goods between sources and sinks.
 - d. These arcs represent the opportunity cost of unused resources or unmet demand.
 - e. These arcs represent a way to balance supply and demand without incurring any additional cost.
9. Slides 7-8 present a more general approach to handling inequality constraints in network flow problems using a dummy node. Explain how this approach differs from the specific method used for the transportation problem in Slides 3-5, and what advantages or disadvantages this more general approach might offer.
- a. The general approach uses a single dummy node regardless of whether supply exceeds demand or vice versa, while the transportation problem approach uses separate dummy nodes for each case.
 - b. The general approach allows for both upper and lower bounds on arc flows, while the transportation problem approach only handles lower bounds.
 - c. The general approach uses a different type of dummy node that can handle multiple sources and sinks simultaneously, while the transportation problem approach uses separate dummy nodes for each source and sink.
 - d. The general approach is applicable to any network flow problem with inequality constraints, while the transportation problem approach is specific to transportation problems.
 - e. The general approach introduces additional constraints to ensure feasibility, while the transportation problem approach relies solely on the dummy node to balance supply and demand.
10. In the transportation problem with inequality constraints (Slides 1-2), if the total supply $\sum_i s_i$ is strictly less than the total demand $\sum_j d_j$, what modification is necessary to transform the problem into a standard transportation problem solvable with equality constraints, and why is this modification needed?
- a. Introduce a dummy source node with a supply equal to $\sum_j d_j - \sum_i s_i$, and connect this dummy node to all demand nodes with zero-cost arcs.
 - b. Introduce a dummy demand node with a demand equal to $\sum_i s_i - \sum_j d_j$, and connect all supply nodes to this dummy node with zero-cost arcs.
 - c. Increase the supply at each source node proportionally to make the total supply equal to the total demand.

- d. Decrease the demand at each demand node proportionally to make the total demand equal to the total supply.
- e. The problem is already solvable as is; no modification is needed.

2 Answers

1. In the transportation problem with inequality constraints (Slides 1-2), if the total supply $\sum_i s_i$ exceeds the total demand $\sum_j d_j$, what modification is necessary to transform the problem into a standard transportation problem solvable with equality constraints, and why is this modification needed?
 - a. Introducing artificial variables to the supply constraints would not address the imbalance between supply and demand. Artificial variables are typically used to find an initial feasible solution in the simplex method, not to handle unbalanced supply and demand in a transportation problem.
 - b. A dummy destination node is introduced to absorb the excess supply. This transforms the inequality constraints ($\sum_j x_{ij} \leq s_i$ and $\sum_i x_{ij} \geq d_j$) into equality constraints, allowing the application of standard transportation algorithms which require balanced supply and demand. Without this modification, the standard algorithms would not function correctly.
 - c. Introducing a dummy source node would not correctly address the imbalance. The excess supply needs to be absorbed, not created.
 - d. Relaxing the equality constraints to inequalities would not solve the problem; it would simply change the nature of the problem, not make it solvable by standard algorithms.
 - e. Proportionally adjusting supply and demand would alter the original problem, potentially leading to an inaccurate solution.
2. Slides 3 and 4 introduce a dummy node to handle inequality constraints in the transportation problem. Explain the role of the additional constraint derived in Slide 4, $\sum_i s_i - \sum_i x_{id} = \sum_j d_j + \sum_j x_{dj}$, in ensuring the feasibility and solvability of the modified problem.
 - a. While the constraint implicitly relates to flow into and out of the dummy node, its primary function is to balance total supply and demand.
 - b. The additional constraint ensures that the total supply, considering the flow to the dummy node, equals the total demand, considering the flow from the dummy node. This creates a balanced transportation problem, which is a necessary condition for the application of standard solution methods. The equality ensures feasibility.
 - c. Non-negativity is enforced separately; this constraint focuses on the balance of supply and demand.
 - d. The linearity of the objective function is not directly affected by this constraint; it is a constraint on the flow variables.
 - e. The constraint is crucial for transforming the problem into a solvable form.
3. In Slide 5, a transportation problem with unbalanced supply and demand is transformed by introducing a dummy node. Explain the economic interpretation of the zero-cost arcs connecting the dummy node to the sources and sinks.
 - a. If there were costs associated with storage or procurement, these costs would be reflected in non-zero arc weights.
 - b. Penalties for unmet supply or demand would be represented by non-zero costs associated with the dummy node.
 - c. The costs of transporting goods between sources and sinks are represented by the costs on the arcs connecting the sources and sinks directly, not the dummy node.
 - d. Opportunity costs are not directly represented in this model; the focus is on the explicit costs of transportation.

- e. The zero cost arcs represent the mathematical mechanism for balancing supply and demand. The dummy node absorbs excess supply or provides for unmet demand without any associated cost. This allows the problem to be solved using standard methods.
4. Slides 7-8 present a more general approach to handling inequality constraints in network flow problems using a dummy node. Explain how this approach differs from the specific method used for the transportation problem in Slides 3-5, and what advantages or disadvantages this more general approach might offer.
 - a. Both approaches use slack variables to convert inequalities to equalities.
 - b. Both approaches can handle both equality and inequality constraints, but the general approach is more flexible in how it handles them.
 - c. Both approaches typically use a single dummy node.
 - d. The general approach (Slides 7-8) is more flexible because it can handle a wider variety of inequality constraints ($\geq$ and $\leq$) in a network flow problem, not just those specific to the transportation problem. The transportation problem approach is a specialized case of the more general method. The general approach might be less efficient for specific problems like the transportation problem, where specialized algorithms exist.
 - e. The computational efficiency depends on the specific problem and algorithms used; the general approach is not inherently less efficient.
 5. In the transportation problem with inequality constraints (Slides 1-2), if the total supply $\sum_i s_i$ is less than the total demand $\sum_j d_j$, how can the problem be modified to ensure feasibility and solvability using standard transportation problem algorithms?
 - a. Answer A is correct. Introducing a dummy source node with a supply equal to the difference between total demand and total supply effectively creates a balanced transportation problem. The zero-cost arcs ensure that the additional supply does not affect the optimal solution's cost. This approach transforms the inequality-constrained problem into an equality-constrained problem solvable by standard algorithms.
 - b. Answer B is incorrect. Proportionally increasing the supply might lead to an unrealistic or infeasible solution, as it alters the original supply constraints.
 - c. Answer C is incorrect. Proportionally decreasing the demand might lead to an unrealistic or infeasible solution, as it alters the original demand constraints.
 - d. Answer D is incorrect. Ignoring the inequality constraints will likely lead to an infeasible solution, as the supply will not be sufficient to meet the demand.
 - e. Answer E is incorrect. While introducing a dummy demand node might seem plausible, it would not address the core issue of insufficient supply to meet the demand. The dummy node would absorb excess supply, but the demand would still exceed the available supply.
 6. In the transportation problem with inequality constraints (Slide 1), if the total supply $\sum_i s_i$ is less than the total demand $\sum_j d_j$, how can the problem be modified to ensure feasibility and solvability using standard transportation problem algorithms?
 - a. A dummy source with supply equal to the unmet demand is introduced. Zero-cost arcs connect this dummy source to all destinations, representing the unmet demand. This ensures that the total supply equals the total demand, allowing the use of standard transportation algorithms.

- b. Proportionally reducing demands is not a valid approach, as it alters the original problem's requirements. The solution would not reflect the true optimal allocation given the original demands.
 - c. Ignoring excess demand leads to an infeasible solution, as the demands cannot be fully met. Standard algorithms require balanced supply and demand.
 - d. Introducing a dummy destination with demand equal to the excess supply is incorrect. This would not address the unmet demand and would lead to an infeasible solution.
 - e. Proportionally increasing supplies is not a valid approach, as it alters the original problem's constraints. The solution would not reflect the true optimal allocation given the original supplies.
7. Slides 3 and 4 introduce a dummy node to handle inequality constraints in the transportation problem. Explain the role of the additional constraint derived in Slide 4, $\sum_i s_i - \sum_i x_{id} = \sum_j d_j + \sum_j x_{dj}$, in ensuring the feasibility and solvability of the modified problem.
- a. The constraint does not directly minimize the total cost; that is the objective function's role.
 - b. The additional constraint balances the total supply and demand, making the problem solvable by standard transportation algorithms which require balanced supply and demand. The left side represents total supply minus the amount not used, and the right side represents total demand plus the amount supplied from the dummy node.
 - c. Non-negativity is ensured by other constraints ($x_{ij}, x_{id}, x_{dj} \geq 0$).
 - d. Supply constraints are handled by the individual source constraints ($\sum_j x_{ij} + x_{id} = s_i$).
 - e. Demand constraints are handled by the individual destination constraints ($\sum_i x_{ij} = d_j + x_{dj}$).
8. In Slide 5, a transportation problem with unbalanced supply and demand is transformed by introducing a dummy node. Explain the economic interpretation of the zero-cost arcs connecting the dummy node to the sources and sinks.
- a. If there were costs associated with storage or procurement, these would be non-zero.
 - b. Penalties for unmet supply or demand would be represented by non-zero costs.
 - c. The costs of transporting goods between sources and sinks are represented by the costs on the arcs between the sources and sinks, not the dummy node.
 - d. While there is an opportunity cost, it is not directly represented by the arcs; the zero cost is a simplification for the mathematical model.
 - e. The zero cost represents the fact that there is no cost associated with either excess supply (from source to dummy) or unmet demand (from dummy to sink). The dummy node is a mathematical construct to balance the problem.
9. Slides 7-8 present a more general approach to handling inequality constraints in network flow problems using a dummy node. Explain how this approach differs from the specific method used for the transportation problem in Slides 3-5, and what advantages or disadvantages this more general approach might offer.
- a. Both approaches use a single dummy node to balance supply and demand.
 - b. Both approaches can handle lower bounds (demand constraints). The general approach can also handle upper bounds, but this is not the primary difference.
 - c. Both approaches use a single dummy node to handle imbalances.

- d. The approach in Slides 7-8 is a more general method applicable to any network flow problem with inequality constraints, whereas Slides 3-5 focus specifically on the transportation problem. The general approach offers broader applicability but might not be as efficient for specific problem structures like the transportation problem.
 - e. Both approaches introduce additional constraints (implicitly or explicitly) to ensure feasibility.
10. In the transportation problem with inequality constraints (Slides 1-2), if the total supply $\sum_i s_i$ is strictly less than the total demand $\sum_j d_j$, what modification is necessary to transform the problem into a standard transportation problem solvable with equality constraints, and why is this modification needed?
- a. Answer A is correct. Since the total supply is less than the total demand, there will be unmet demand. Introducing a dummy source node with a supply equal to the difference between total demand and total supply allows us to model the unmet demand. Connecting this dummy node to all demand nodes with zero-cost arcs represents the fact that this unmet demand incurs no cost. This transforms the problem into a balanced transportation problem with equality constraints, which can be solved using standard algorithms.
 - b. Answer B is incorrect. Introducing a dummy demand node with a demand equal to the difference between total supply and total demand would not solve the problem. The problem is that there is not enough supply to meet the demand. A dummy demand node would not address this issue.
 - c. Answer C is incorrect. Proportionally increasing the supply at each source node would change the original problem statement and is not a valid approach to solving the inequality constraint problem.
 - d. Answer D is incorrect. Proportionally decreasing the demand at each demand node would change the original problem statement and is not a valid approach to solving the inequality constraint problem.
 - e. Answer E is incorrect. Standard transportation problem algorithms require equality constraints (supply equals demand). The inequality constraints make the problem unsolvable using standard methods without modification.